

TRAINING @ FORREST.ONE

Session 2 of 12

Network Fundamentals + CompTIA Security+ SY0-701

Switching, VLANs & Network Architectures

Devices, routes and segments: how switches and routers move traffic, how routing protocols and address translation choose paths, how VLANs, trunks and spanning tree segment

NETWORK and **SWITCHING** devices **ROUTING** and **VLANs** can decide what an attacker can reach.



SESSION OBJECTIVES

By the end of this session you will be able to

- **Compare switching with routing** frames by MAC address against packets by IP address
- **Describe the security appliances** firewalls, intrusion detection and intrusion prevention
- **Explain the traffic shapers** load balancers, forward proxies and reverse proxies
- **Compare the storage models** network attached storage against a storage area network
- **Define the cloud terms and models** virtualisation, private networks, security groups, gateways
- **Compare the routing protocols** static, distance vector, link state and path vector
- **Explain route selection** administrative distance, then the metric, then the longest match
- **Describe translation and redundancy** NAT, PAT, first hop redundancy, VLANs and trunks



CHAPTER 1 · INFRASTRUCTURE DEVICES

Network Infrastructure Devices

- **Switches and routers** layer 2 frame forwarding against layer 3 path selection
- **Security appliances** firewalls, intrusion detection and intrusion prevention
- **Traffic shaping devices** load balancers, forward proxies and reverse proxies
- **Storage and wireless** network attached storage, storage area networks, access points
- **Cloud networking** virtualised network functions, private clouds, service models

The thread

Every device is also a security boundary. Ask of each one, who does this protect, and from whom? Detection alone is not defence.



CHAPTER 1 · INFRASTRUCTURE DEVICES

Switches forward frames, routers forward packets

- **Switch** a layer 2 device, forwarding frames by MAC address
- **MAC address** 48 bits as six hex pairs; the first three octets are the OUI
- **Learning** the table is built from the source address on every arriving frame
- **Scope** a switching decision stays inside one network or broadcast domain
- **Router** a layer 3 device, forwarding packets by IP address between networks
- **Routing table** a next hop per destination, learned by protocols such as OSPF and BGP

The distinction that matters

Switching is local and joins devices; routing joins networks. A capture shows the hardware addresses of the devices that spoke on that segment, even when the traffic later crossed a router.



CHAPTER 1 · INFRASTRUCTURE DEVICES

The switch forwarding table

1

Learn

record the source MAC and the port it arrived on

2

Look up

find the destination MAC in the table

3

Forward

out of the one port that holds the destination

4

Flood

unknown unicast goes to every port but the source

5

Broadcast

always flooded, which is why a loop is fatal

MAC flooding

Fill the table with bogus source addresses and the switch degrades to hub mode, so every frame becomes visible to the attacker. That is why port security limits how many addresses may appear on one port.



CHAPTER 1 · INFRASTRUCTURE DEVICES

Security appliances: detect, prevent, filter

- **Firewall** filters on rules built from addresses, ports, protocols and content
- **Placement** at the perimeter, and internally between zones
- **Intrusion detection** monitors passively and raises an alert, but never blocks
- **Intrusion prevention** sits inline and can drop the packet or reset the connection
- **Sensors** a network sensor watches a segment; a host sensor watches one machine
- **Defence in depth** all three layers together, because one will be bypassed

Say it plainly

The firewall refuses by rule, detection reports after the fact, and prevention stops it in line. An alert with nobody watching it is not a control.



CHAPTER 1 · INFRASTRUCTURE DEVICES

Load balancers and proxies

- **Load balancer** spreads requests across a pool so no server is overwhelmed
- **Round robin** each request goes to the next server in turn
- **Least connections** each request goes to the server with the fewest active sessions
- **Forward proxy** sits in front of the clients, hides them and filters outbound requests
- **Reverse proxy** sits in front of the servers, caches content and terminates TLS
- **Availability** one failed server no longer takes the service down

Which side is protected

A forward proxy protects the client side of the conversation and a reverse proxy protects the server side. Both hide one side from the other.



CHAPTER 1 · INFRASTRUCTURE DEVICES

Storage and wireless hardware

Device	Role	Protocol or standard
Network attached storage	File level sharing over the network	NFS and SMB
Storage area network	Block level storage for servers	Fibre Channel
Access point	Bridges wireless clients onto the wired network	802.11ac and 802.11ax
Wireless controller	Configures many access points centrally	One policy interface
Wireless bridge	Joins two segments with no cable	Point to point

Pairing rule

NAS is file level over the network with NFS and SMB; a SAN is block level with Fibre Channel and looks to the server like a local disk. Learn the pairings, because the exam asks which one a server actually needs.



DISCUSSION

A firewall, an intrusion detection system and an intrusion prevention system all see the same traffic. Which one stops an attack, and why can you not rely on it alone?

Prevention blocks inline, detection only reports, and the firewall refuses by rule. Defence in depth assumes one of the three will be bypassed or misconfigured.

An IPS is only as good as its signatures and its tuning; a firewall rule set is only as good as its last review. Ask the room who actually reads the alerts.



CHAPTER 2 · CLOUD & ROUTING

Cloud Networking and Routing Foundations

- **Cloud computing** replaces dedicated appliances with software, and moves the boundary
- **Routing** begins with a static route typed by hand
- **Then** a dynamic protocol that learns the topology for itself
- **The question to keep asking** what breaks if this mechanism is misconfigured?

The thread continues

Watch where the attack surface sits, because it moves with every design choice. Virtual appliances put the management plane and the hypervisor in the attacker's sights, and every routing option decides who reaches what.



CHAPTER 2 · CLOUD & ROUTING

Network functions virtualisation

- **Virtualisation** firewalls, routers and load balancers become software
- **Platform** they run on ordinary servers or virtual machines
- **Benefits** lower cost, faster deployment and easy scaling
- **Speed** a virtual firewall deploys in a moment, not after a procurement cycle
- **New targets** the management plane and the hypervisor
- **The trade** the same function, but the security boundary has moved

What moved

The appliance left the rack, so the patch cycle, the console access and the configuration file are now software problems that somebody must own.



CHAPTER 2 · CLOUD & ROUTING

Private clouds, security groups and gateways

- **Virtual private cloud** a logically isolated network inside a public cloud
- **You define** the address ranges, the subnets and the route tables
- **Security groups** stateful firewalls on a resource, so replies are allowed automatically
- **Network ACLs** instead filter at the subnet boundary
- **VPN gateway** an encrypted tunnel back to the office
- **Direct connection** private, and avoids the public internet entirely

Design rule

Default to deny, then open only what the service actually needs. Every open rule is a decision somebody will have to justify later.



CHAPTER 2 · CLOUD & ROUTING

Deployment and service models

Model	Who manages what	Typical use
Public cloud	Provider owns the infrastructure	Scale, pay as you go
Private cloud	One organisation, on site or hosted	Strict compliance needs
Hybrid cloud	Sensitive data private, the rest public	Control with lower cost
Software as a service	Provider runs the whole application	Mail and office suites
Infrastructure as a service	You run operating systems and data	Rented compute and storage
Platform as a service	You build only the application	Application development

Where the responsibility stops

Service models shift the boundary of responsibility, never the whole of it. On software as a service you still manage your data and your users; on infrastructure as a service you also patch the operating system.



CHAPTER 2 · CLOUD & ROUTING

Static routing

- **Static route** typed by an administrator and never changed on its own
- **Good for** small networks, stub networks and default routes
- **Stub network** one way in and one way out
- **Default route** catches every destination the table does not list
- **Administrative distance** 1, so it is trusted above anything learned dynamically
- **The cost** it does not adapt, so a failed link stays in the table

Predictable, until it is not

A static route gives control and no surprises until the topology moves, and then traffic is sent into a dead end that the table still believes in.



CHAPTER 2 · CLOUD & ROUTING

Dynamic routing and distance vector

- **Dynamic protocol** learns by exchanging information with its neighbours
- **Distance vector** a router advertises its whole table: routing by rumour
- **RIP** Routing Information Protocol measures a path in hop count
- **The limits** 15 hops is the maximum usable count, and 16 means unreachable
- **Convergence** slow convergence brings routing loops and black holes
- **Defences** split horizon, route poisoning and hold down timers

Why 16 exists

The hop limit is how RIP stops a route circling for ever. Neighbours share what they know until the network converges, and while they disagree, traffic disappears.



CHAPTER 2 · CLOUD & ROUTING

Open Shortest Path First

- **OSPF** the open standard link state protocol
- **Link state advertisement** each router describes its own links to every other router
- **Link state database** every router holds the same one, so the same map of the network
- **Shortest path first** each router runs the calculation with itself as the root
- **Cost** follows bandwidth, so a lower cost is the preferred link
- **Areas** area 0 is the backbone, and every other area attaches to it

The link state idea

Distance vector routers trust rumour; link state routers each hold the whole map and then work out their own best path from it.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

Route Selection, Translation and Inter-VLAN Routing

- **Choosing** administrative distance ranks the source a route came from
- **Translating** we rewrite the addresses themselves as traffic crosses a boundary
- **Protecting** first hop redundancy makes the default gateway survive a failure
- **Carrying** one physical link carries every VLAN: router on a stick

Why this chapter exists

Each of these is a control you may be asked to justify, and each fails in a way that looks like something else at first. Misconfiguration here is one of the most common causes of a real outage.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

EIGRP and BGP

- **EIGRP** advanced distance vector, and Cisco proprietary in origin
- **Feasible successor** the diffusing update algorithm keeps a backup path ready
- **Recovery** failover to that backup needs no recalculation of the network
- **Metrics** bandwidth, delay, load and reliability, not hop count alone
- **BGP** the path vector protocol of the internet
- **Selection** attributes and policy, such as the autonomous system path length

Two worlds

EIGRP is fast inside the enterprise; BGP carries the internet and is the protocol behind route hijacks. Both must be secured and monitored.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

Administrative distance decides, then the metric

Route source	Administrative distance	Trust
Connected route	0	Highest
Static route	1	Very high
EIGRP	90	High
OSPF	110	Medium
RIP	120	Low

Distances first, then the metric

A router often learns the same destination from more than one source, and administrative distance settles the argument because the lowest value wins. Once the source is chosen, its metrics pick between paths, and the longest match rule means the most specific route wins overall.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

Network address translation

- **NAT** rewrites addresses in the packet header as traffic crosses a boundary
- **Reason one** to slow the exhaustion of the IPv4 address pool
- **Reason two** to hide the internal layout from the outside
- **Static NAT** one private address to one public address, for a published server
- **Dynamic NAT** an address taken from a pool, first come first served
- **Vocabulary** inside and outside give position; local and global give scope

Log the translations

A translation table is evidence and a troubleshooting tool at the same time. Without it, an outbound connection cannot be traced back to a workstation.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

Port address translation

- **PAT** maps many private addresses onto one public address
- **The trick** every session is given a unique port number
- **Translation table** tracks every active connection
- **The result** two hosts appear as one public address on two different ports
- **Inbound services** need a port forwarding rule, because the router cannot know who asked
- **In the wild** most home routers run PAT by default

The everyday form

Port address translation is the version of NAT that hides a whole office behind a single public address, told apart by port number alone.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

First hop redundancy protocols

- **The problem** if the default gateway dies, everything behind it loses the outside world
- **The fix** a shared virtual IP address that floats between routers
- **HSRP** Hot Standby Router Protocol keeps one active router and one standby
- **VRRP** the open standard version, with a master and several backups
- **GLBP** goes further and shares the forwarding load between routers
- **Preemption** hands the role back to the best router

Transparent failover

Clients never learn that the gateway changed underneath them, because the address they were configured with is still answering. Test the failover before you need it.



CHAPTER 3 · ROUTE SELECTION & TRANSLATION

Subinterfaces and router on a stick

- **Subinterface** a logical interface built on one physical router port
- **Each one** serves a single VLAN
- **The design** the router reaches every VLAN over one shared trunk
- **Tagging** each subinterface carries the 802.1Q tag for its own VLAN
- **Result** the router routes between the VLANs and each gets a gateway address
- **The cost** all VLANs share one physical link, so it can become the bottleneck

One link, several gateways

Router on a stick is cheap and easy to draw, and it puts every VLAN on one conversation. A busy campus will feel that link.



BREAK
10 MINUTES

Back soon: VLANs, trunking, spanning tree, wireless deployment and the room itself.



CHAPTER 4 · VLANS & SPANNING TREE

VLANS, Switching and Spanning Tree

- **VLANS** split one switch into logical broadcast domains, the core containment control
- **Trunks and tags** how many VLANs cross one link, and how they stay apart
- **Port security** and the guard features that protect an access port
- **Spanning tree** one loop free path, with redundancy held in reserve
- **Frames** why the maximum size breaks networks quietly when it is wrong

Why these five

Every mechanism in this chapter exists for availability or for containment, and every one of them fails in a way that looks like something else at first. That is what makes this chapter worth your attention.



CHAPTER 4 · VLANS & SPANNING TREE

VLAN fundamentals

- **VLAN** a logical broadcast domain built inside the switching hardware
- **Without one** every device on the switch hears every broadcast
- **Membership** by port, by hardware address or by protocol
- **VLAN 1** the default; the native VLAN is the one carrying untagged frames
- **Management VLAN** keeps administration traffic away from users
- **The payoff** fewer broadcasts, a smaller blast radius, no easy lateral movement

Segmentation is a control

A guest VLAN keeps visitors away from internal resources, and naming your VLANs properly makes the design document itself.



CHAPTER 4 · VLANS & SPANNING TREE

Trunking and tagging

- **Trunk port** carries many VLANs between switches
- **802.1Q** inserts a four byte tag into the Ethernet frame
- **The VID** 12 bits, giving just over four thousand usable VLANs
- **Receiving switch** reads the tag and keeps the traffic in the right VLAN
- **Native VLAN** crosses the trunk untagged, so a mismatch can leak between VLANs
- **Access port** carries a single VLAN to one endpoint

VLAN hopping

Two methods: switch spoofing, where the attacker negotiates a trunk, and double tagging, where the native VLAN is spoofed. Tag every trunk consistently and treat an access port as access only.



CHAPTER 4 · VLANS & SPANNING TREE

A practical VLAN design

VLAN	Name	Purpose
10	MANAGEMENT	Switch and infrastructure administration
20	SERVERS	Internal servers, never user traffic
30	WORKSTATIONS	User workstations
40	DMZ	Public facing services
999	BLACKHOLE	Unused quarantine VLAN for dead ports

Key rule

VLAN 1 is the default native VLAN and a known attack surface. Move it to an unused ID, prune unused VLANs from every trunk, and set trunks to explicit allowed lists. That is what blunts VLAN hopping, where an attacker spoofs a trunk or double tags a frame on the native VLAN to jump segments.



CHAPTER 4 · VLANS & SPANNING TREE

Switch interface configuration

- **Speed and duplex** must agree at both ends of a link
- **A mismatch** brings collisions and very poor throughput
- **Port security** limits how many hardware addresses may appear on a port
- **Violations** can drop the traffic, log it, or shut the port down
- **PortFast** skips the listening states on a port that leads to one endpoint
- **BPDU Guard** shuts a port that receives a bridge protocol data unit

The quiet access port

A port that leads to one workstation should never speak spanning tree. BPDU Guard is how you find out that somebody plugged a switch into it.



CHAPTER 4 · VLANS & SPANNING TREE

Loops and the root bridge

- **Redundancy** two links between switches create a layer 2 loop
- **Broadcast storm** frames circulate, and the network is consumed in seconds
- **The MAC table** destabilises as one address appears on two ports
- **STP** builds one loop free logical topology by blocking redundant ports
- **Root bridge** the lowest bridge identifier wins: priority plus MAC address
- **Root port** each switch picks its best port toward the root; the rest block

One logical path

One logical path, no loops, with the redundant link held in reserve until the forwarding path fails.



CHAPTER 4 · VLANS & SPANNING TREE

STP port states and convergence

State	Forwards?	Role
Blocking	No	Listens for BPDUs only, holds the link in reserve
Listening	No	Transitional, still learning the topology
Learning	No	Builds the MAC table but forwards nothing
Forwarding	Yes	Normal operation
Disabled	No	Administratively down

Convergence

Blocking, listening and learning all mean no data. Traditional STP takes thirty to fifty seconds to converge after a topology change; rapid spanning tree does it in seconds. Know which one your campus runs.



CHAPTER 4 · VLANS & SPANNING TREE

Rapid and multiple spanning tree

- **RSTP** 802.1w converges in seconds rather than tens of seconds
- **New roles** it adds the alternate and backup port roles
- **New states** discarding replaces the blocking and listening states
- **MSTP** 802.1s maps groups of VLANs to instances, so redundant links get used
- **Per VLAN STP** instead gives every VLAN its own tree
- **Root placement** choose the root bridge deliberately, never leave it to chance

Where you want the root

The root bridge should be the switch at the centre of your design, not the one that happened to have the lowest hardware address on the day.



CHAPTER 4 · VLANS & SPANNING TREE

STP attacks and the guard features

Threat	Feature	Effect
Attacker becomes the root bridge	Root Guard	Port cannot become a root port
BPDUs on an access port	BPDU Guard	Port shuts down on receipt
Rogue switch insertion	Port security	Limits the MACs allowed per port
Flooding the MAC table	Port security	Stops the switch degrading to hub mode

Why it matters

A rogue root bridge forces all traffic through the attacker's switch, which is invisible interception rather than an outage. Root Guard and BPDU Guard are the two lines of defence, and port security blunts MAC flooding at the same time.



DISCUSSION

Two switches, equal priority. Which becomes root, and what converges?

The tiebreaker is the MAC address: the lower one wins. Then each non-root switch picks its root port by path cost to the root.

The gigabit link at cost four forwards and the hundred megabit link at cost nineteen blocks. If the gigabit link fails, the blocked port transitions and the network reconverges: seconds on RSTP, thirty to fifty seconds on classic STP.



Maximum transmission unit and jumbo frames

- **Standard MTU** 1500 bytes for Ethernet
- **Larger packet** must be fragmented, or dropped if fragmentation is refused
- **The signal** an ICMP message returns to the sender when a fragment is refused
- **Jumbo frames** raise the limit to about 9000 bytes
- **Why bother** they cut header overhead in data centres and storage networks
- **The trap** every device in the path must agree, or packets simply vanish

A quiet failure

An MTU mismatch is a classic cause of strange, partial failures: small requests that work, large transfers that stall, and no obvious error anywhere.



QUIZ

Activity: design the segmentation

- Which VLAN should host the web server facing the internet?
- Which VLAN should never carry user traffic?
- Why isolate management traffic from workstations?
- What risk does the default native VLAN create?
- Name two attacks VLAN segmentation prevents.

The DMZ hosts public services, the blackhole swallows unused access ports, and management should never share a domain with users. Think microsegmentation: isolate the sensitive workloads, not just whole departments.



SESSION 2 · SEGMENTATION & ZONES

From segments to security zones

- **Segmentation** VLANs and subnets become security zones
- **DMZ** public facing services, isolated from the internal network
- **DMZ firewalls** an external and an internal firewall, often with a proxy between
- **Blast radius** a compromise stays inside its own segment
- **Lateral movement** zones block east-west spread across the campus
- **Campus and IoT** access connects, distribution enforces policy, unmanaged devices isolated

The exam line

Architecture is a control, not plumbing. When a question asks how to contain a breach, the answer is nearly always segmentation.



CHAPTER 5 · WIRELESS & PHYSICAL

Wireless and Physical Deployment

- **Frequencies** the bands and channel widths, and what each one is good for
- **Names and modes** the identifiers a wireless network uses
- **Encryption** a history of what happens when a control is broken early
- **Authentication** enterprise sign in, and guest design
- **The room** the rack, the cable and the physical access

The principle returns

Notice how often the same idea comes back: separate what must not touch. And notice how often physical access decides the final outcome.



Channels and frequencies

- **2.4 GHz** eleven channels, and only 1, 6 and 11 do not overlap
- **5 GHz** up to twenty three non overlapping channels, so busy offices prefer it
- **6 GHz** used by Wi-Fi 6E, and adds far more spectrum again
- **Lower frequencies** travel further and pass through walls better
- **Higher frequencies** carry more data over shorter distances
- **Channel width** wider channels raise throughput and consume the space around them

Choose for the job

Pick the band for the requirement, not for the marketing number. Twenty megahertz is the steady default, and adjacent access points must not overlap.



Network names and modes

- **SSID** the service set identifier, the network name clients see
- **Infrastructure mode** sends all traffic through an access point
- **Ad hoc mode** devices talk directly, with weaker security and no central control
- **BSSID** the basic service set identifier names a single access point
- **ESSID** the extended service set identifier lets clients roam between access points
- **Hiding the name** stops nothing; client isolation keeps users apart

Roaming

Roaming needs one shared name and separate radios behind it, with the controller setting power and channel so the client moves when it should.



Wireless encryption

- **WEP** wired equivalent privacy: short keys and a weak schedule, broken quickly
- **WPA** added the temporal key integrity protocol as a stopgap
- **WPA2** replaced that with AES, and is still the common baseline
- **WPA3** adds simultaneous authentication of equals, resisting offline cracking
- **Personal mode** one pre shared key for everybody
- **Enterprise mode** 802.1X with a central authentication server

The rule

Retire WEP entirely. Run WPA2 or WPA3 with AES, and move to enterprise mode wherever you have an authentication server to point at.



CHAPTER 5 · WIRELESS & PHYSICAL

Enterprise authentication and guest networks

- **802.1X** authenticates a device before it joins the network
- **Authentication server** checks credentials centrally, so the access point does not decide
- **EAP** the extensible authentication protocol carries the exchange
- **Protected EAP** hides that exchange inside a TLS tunnel
- **EAP TLS** uses certificates on both the client and the server
- **Guest network** its own name, its own VLAN, a bandwidth limit and clear rules

Guests get the internet

Guests get the internet and nothing else, and they accept an acceptable use policy before they get even that. Survey before you mount.



CHAPTER 5 · WIRELESS & PHYSICAL

Access points and antennas

- **Standalone access point** configured on its own, fine for a small site
- **Controller based** managed centrally, with power and channel set for every unit
- **Consistency** the controller keeps policy identical across the estate
- **Omnidirectional** radiates all around, for open areas
- **Directional** a Yagi focuses a beam for a long point to point link
- **Gain and survey** more gain means more reach; a heat map shows real coverage

Measure, do not guess

Place access points by measurement, not by eye. A survey before installation is cheaper than a second visit after the complaints.



CHAPTER 5 · WIRELESS & PHYSICAL

Installation, power and environment

- **Rack** a standard equipment rack is nineteen inches wide
- **Heights** one rack unit is 1.75 inches of vertical space
- **Labelling** label every cable and document the patch panel
- **Patch panel** protects the switch ports and eases every change
- **Physical security** lockable racks and a controlled room
- **Cooling and environment** hot and cold aisles, a UPS, and no water, heat or dust

The bottom line

Physical access defeats every logical control you own. Label it, or you will lose it, and troubleshooting becomes guesswork.



QUIZ

Check your understanding

- Open Quiz_Session_02.html in your browser.
- 20 questions, immediate feedback, and a topic tag on every question.
- Read each explanation: the distractors teach as much as the answer.
- Score 70% or better before you move on to Session 3.

Every question is tagged with its banked topic area, covering infrastructure devices, VLANs and trunking, spanning tree and network architectures.



KEY TAKEAWAYS

- **Switches forward frames** routers forward packets.
- **Each appliance does one job** refuse, report, block.
- **Cloud moves the boundary** software functions, security groups.
- **Distance ranks the source** the metric picks the path.
- **The longest match wins first** the most specific route.
- **Translation conserves addresses** NAT and PAT, told apart by port.
- **VLANs contain, trunks carry tags** spanning tree keeps one path.
- **Wireless means WPA2 or WPA3** with AES and a guest VLAN.



REVISION CHECKLIST

Before next session

- **Retake the quiz** until you score 70% or better without guessing
- **Redraw one design** a VLAN plan, a distance table, or a wireless policy
- **Trace an election** two switches, two links: pick the root bridge and the blocked port
- **Read every device twice** once as plumbing, once as a boundary
- **Read ahead** Session 3: DNS, DHCP, NTP and SSH, and how each is attacked



NEXT SESSION

Session 3 · Network Services & the Attack Surface

- DNS: hierarchy, record types, poisoning and DNSSEC
- DHCP: DORA, rogue servers, starvation and snooping
- NTP and SSH: time integrity, and the encrypted management channel
- Every service is a service, and every service is a target

